

RX Lens Pricing and Configuration Specification

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1. Purpose and scope

This document defines the lens options offered on the website and the exact rules used to calculate the RX lens add-on price for sunglass frames. It is intended for internal use (product, development, and operations).

2. Configuration overview

Customers configure prescription lenses through the following decision steps:

2.1 Lens type (mutually exclusive)

- Clear (Mono RX): clear prescription lenses without tint.
- Tinted (Mono RX): clear prescription lenses with mandatory tint service (Full Tint or Gradient).
- Photochromic (Solis II): lenses that darken outdoors and become clear indoors.
- Polarized (NuPolar): polarized sun lenses that reduce glare.

2.2 Lens index

- 1.56 (standard thickness)
- 1.60 (thinner)
- 1.67 (thinnest among offered options)

2.3 Coating (one selection only)

- UC: Uncoated (available only for Clear).
- UV Protection: Clarus Sericum UV.
- Blue PRO: blue-light protective anti-reflective coating.

2.4 Color and tint selections

- Tint color choices affect production instructions and preview only; they do not change price.
- Tint type affects price (Full Tint vs Gradient).

3. Supported options and business rules

3.1 Supported indexes by lens type

Lens type	Supported indexes
Clear (Mono RX)	1.56, 1.60, 1.67
Tinted (Mono RX)	1.56, 1.60, 1.67
Photochromic (Solis II)	1.56, 1.60, 1.67
Polarized (NuPolar)	1.60, 1.67 (1.56 is not offered)

3.2 Allowed coatings by lens type

Lens type	Allowed coatings
Clear (Mono RX)	UC, Blue PRO
Tinted (Mono RX)	UV Protection (Clarus Sericum UV), Blue PRO
Photochromic (Solis II)	UV Protection (Clarus Sericum UV), Blue PRO
Polarized (NuPolar)	UV Protection (Clarus Sericum UV), Blue PRO

Operational note: Blue PRO is available for Tinted, Photochromic, and Polarized lenses as an alternative to UV Protection. In the current supplier price list, Blue PRO and Clarus Sericum UV are priced the same for those lens families and indexes, so Blue PRO does not add an extra charge versus UV Protection (the UI may display “Included”). UC is intentionally not offered for Tinted/Photochromic/Polarized to avoid high reflections and to standardize sun-lens orders.

Auto-corrections: if the user selects an invalid combination (for example Polarized + 1.56 index, or Tinted + UC coating), the configurator automatically switches to the nearest valid option (first supported index and first allowed coating).

4. Tinting options (Tinted lens type only)

Tinting is available only when the lens type is set to Tinted (Mono RX).

4.1 Tint types and fees (per pair)

Tint type	Description	Fee (per pair)
Full Tint - Catalog Colors	Solid tint across the entire lens.	€6.00
Gradient Tint	Darker at the top, fading to lighter at the bottom.	€12.00

4.2 Allowed tint colors

Tint color	Availability
Brown	Allowed for Full Tint and Gradient
Grey	Allowed for Full Tint and Gradient
Green	Allowed for Full Tint and Gradient

4.3 Full tint shade levels (metadata only)

Shade levels are stored as production instructions. They do not change the tint fee.

Color	Available shade levels (solid tint)
Brown	15%, 50%, 85%
Grey	15%, 70%, 85%
Green	15%, 30%, 85%

4.4 Gradient recipes (fixed, metadata only)

Gradient recipes are fixed targets and are not user-editable. They are stored as production instructions.

Color	Gradient recipe (top → bottom)
Grey	30 → 0
Brown	50 → 0
Green	90 → 15

5. Photochromic (Solis II) color options

Option	Allowed values
Photochromic color	Brown, Grey

Note: Photochromic behavior is inherent to the Solis II lens. Optional mirror-style finishes are not part of the current offering.

6. Polarized (NuPolar) color options

Option	Allowed values
Polarized color	Brown, Grey, Green

Operational note: confirm color availability with the lab for the chosen NuPolar index/SKU set. Website options must match production capability.

7. Base lens pricing

Base lens prices are sourced from the BOD price list (2025). Prices in the tables below are per single lens. The website calculates a pair price by multiplying by 2.

7.1 Mono RX (Clear and Tinted base) - prices per single lens

Index	UC	Blue PRO	Clarus Sericum UV
1.56	8.49	14.49	14.49
1.60	16.98	22.98	22.98
1.67	25.07	31.07	31.07

7.2 Solis II Photochromic - prices per single lens

Index	UC	Blue PRO	Clarus Sericum UV
1.56	13.53	19.53	19.53
1.60	24.66	30.66	30.66
1.67	32.90	38.90	38.90

7.3 NuPolar Polarized - prices per single lens

Index	UC	Blue PRO	Clarus Sericum UV
1.60	35.98	41.98	41.98
1.67	58.62	64.62	64.62

7.4 Quick reference - base prices per pair (2 lenses)

These values are the per-pair equivalents of the single-lens tables above (pair = 2 × single).

Lens family	Index	UC (pair)	Blue PRO (pair)	Clarus Sericum UV (pair)
Mono RX	1.56	16.98	28.98	28.98
Mono RX	1.60	33.96	45.96	45.96
Mono RX	1.67	50.14	62.14	62.14
Solis II	1.56	27.06	39.06	39.06
Solis II	1.60	49.32	61.32	61.32
Solis II	1.67	65.80	77.80	77.80
NuPolar	1.60	71.96	83.96	83.96
NuPolar	1.67	117.24	129.24	129.24

Business rule reminder: UC is not offered for Tinted, Photochromic, or Polarized lens types. UV Protection and Blue PRO are used for those lens types.

8. Pricing formulas

8.1 Base pair price

$\text{BasePairPrice} = 2 \times \text{SingleLensPrice}(\text{lensType}, \text{index}, \text{coating})$

8.2 Tint fee (Tinted lens type only)

$\text{TintFeePair} = 0$ for all lens types except Tinted.

If $\text{lensType} = \text{Tinted}$: TintFee (per pair) is €6.00 (Full Tint) or €12.00 (Gradient).

8.3 Lens pair total (before edging/profit/VAT)

$\text{LensPairTotal} = \text{BasePairPrice} + \text{TintFeePair}$

8.4 Coating label logic ("Included" vs "+€X")

To label coating options in the UI, compute the price difference (delta) compared to the cheapest allowed coating for the chosen lens type and index.

$\text{CoatingDeltaPair} = \text{BasePairPrice}(\text{selectedCoating}) - \min(\text{BasePairPrice}(\text{allowedCoatings}))$

If $\text{CoatingDeltaPair} = 0$: show "Included". Otherwise show "+€X per pair".

9. Worked examples by index (1.56 / 1.60 / 1.67)

This section provides step-by-step price calculations for each supported index. Base lens prices are taken per single lens from the supplier price list; the website calculates the pair base as (2 × single-lens price). Tint service fees are applied per pair.

Tinted coating note (why Blue PRO may appear “included”): For Tinted Mono RX, Solis II, and NuPolar in the current supplier price list, Blue PRO and Clarus Sericum UV have the same price at the same index. Therefore selecting Blue PRO does not change the price versus UV Protection for those lens types; the UI may label both as “Included” (price delta = €0). For Clear lenses, UC is cheaper, so Blue PRO shows a surcharge relative to UC.

9.1 Index 1.56

Selection	Single-lens base (€)	Pair base (2×) (€)	Tint fee (pair) (€)	Total formula	Result (pair) (€)
Clear (Mono RX) + UC	8.49	16.98	0.00	$(2 \times 8.49) + 0.00 = 16.98$	16.98
Clear (Mono RX) + Blue PRO	14.49	28.98	0.00	$(2 \times 14.49) + 0.00 = 28.98$	28.98
Tinted (Mono RX) + (UV Protection or Blue PRO) + Full Tint (Catalog)	14.49	28.98	6.00	$(2 \times 14.49) + 6.00 = 34.98$	34.98
Tinted (Mono RX) + (UV Protection or Blue PRO) + Gradient Tint	14.49	28.98	12.00	$(2 \times 14.49) + 12.00 = 40.98$	40.98
Photochromic (Solis II) + (UV Protection or Blue PRO)	19.53	39.06	0.00	$(2 \times 19.53) + 0.00 = 39.06$	39.06
Polarized (NuPolar) + (UV Protection or Blue PRO)	N/A	N/A	N/A	Not offered for this index	N/A

9.2 Index 1.60

Selection	Single-lens base (€)	Pair base (2×) (€)	Tint fee (pair) (€)	Total formula	Result (pair) (€)
Clear (Mono RX) + UC	16.98	33.96	0.00	$(2 \times 16.98) + 0.00 = 33.96$	33.96
Clear (Mono RX) + Blue PRO	22.98	45.96	0.00	$(2 \times 22.98) + 0.00 = 45.96$	45.96
Tinted (Mono RX) + (UV Protection or	22.98	45.96	6.00	$(2 \times 22.98) + 6.00 = 51.96$	51.96

Blue PRO) + Full Tint (Catalog)					
Tinted (Mono RX) + (UV Protection or Blue PRO) + Gradient Tint	22.98	45.96	12.00	$(2 \times 22.98) + 12.00 = 57.96$	57.96
Photochromic (Solis II) + (UV Protection or Blue PRO)	30.66	61.32	0.00	$(2 \times 30.66) + 0.00 = 61.32$	61.32
Polarized (NuPolar) + (UV Protection or Blue PRO)	41.98	83.96	0.00	$(2 \times 41.98) + 0.00 = 83.96$	83.96

9.3 Index 1.67

Selection	Single-lens base (€)	Pair base (2×) (€)	Tint fee (pair) (€)	Total formula	Result (pair) (€)
Clear (Mono RX) + UC	25.07	50.14	0.00	$(2 \times 25.07) + 0.00 = 50.14$	50.14
Clear (Mono RX) + Blue PRO	31.07	62.14	0.00	$(2 \times 31.07) + 0.00 = 62.14$	62.14
Tinted (Mono RX) + (UV Protection or Blue PRO) + Full Tint (Catalog)	31.07	62.14	6.00	$(2 \times 31.07) + 6.00 = 68.14$	68.14
Tinted (Mono RX) + (UV Protection or Blue PRO) + Gradient Tint	31.07	62.14	12.00	$(2 \times 31.07) + 12.00 = 74.14$	74.14
Photochromic (Solis II) + (UV Protection or Blue PRO)	38.90	77.80	0.00	$(2 \times 38.90) + 0.00 = 77.80$	77.80
Polarized (NuPolar) + (UV Protection or Blue PRO)	64.62	129.24	0.00	$(2 \times 64.62) + 0.00 = 129.24$	129.24

10. Order metadata (recommended fields)

Store the following values in the order/cart line metadata so the lab receives unambiguous production instructions:

- lensType (CLEAR / TINTED / PHOTOCHROMIC_SOLIS / POLARIZED_NUPOLAR)
- lensIndex (1.56 / 1.60 / 1.67)
- coating (UC / SERICUM_UV / BLUE_PRO)
- lensPairTotal (computed)
- If TINTED: tintType (FULL_TINT_CATALOG or GRADIENT), tintColor (Brown/Grey/Green)
- If Full Tint: tintShadePercent (15/30/50/70/85 as applicable)
- If Gradient: tintRecipe (Grey 30->0, Brown 50->0, Green 90->15)
- If PHOTOCHROMIC: photochromicColor (Brown/Grey)
- If POLARIZED: polarizedColor (Brown/Grey/Green)

11. UI swatch reference (recommended)

These hex values are recommended for UI color buttons and previews only; they are not lab dye codes.

Color name	HEX (recommended for UI)
Brown	#B2561D
Grey	#6D7481
Green	#2B8244